

Sai Subrahmanyam Mahadasa

Jersey City, NJ | +1 660-528-5209 | saimahadasa1999@gmail.com | [LinkedIn](#) | [GitHub](#)
Senior Full Stack Developer | 5+ Years | React.js · Gen AI · TypeScript · Python · Django · AWS

Work Experience

Perky (Perspective Partners LLC)

Senior Software Developer

Fairport, NY | Hybrid

Mar 2024 – Mar 2026

- Designed a multi-agent AI pipeline using **LangGraph**, **FastAPI**, **OpenAI**, and AWS Bedrock with **ChromaDB** vector embeddings, handling 500+ concurrent users at sub-2s response times with built-in fallback handling for low-confidence outputs.
 - Developed LLM-powered applications using **LangChain** and **AWS Bedrock**, including conversational assistants, document Q&A systems, and health analysis tools with custom prompt templates, memory management, and streaming responses, enabling hybrid cloud and on-device inference via OpenAI and Ollama (Gemma3:1b).
 - Developed responsive, accessible UI using **React.js**, **Next.js** and **TypeScript** dashboards for a health benefits platform using Redux and RTK Query, cutting page load times from 3.5s to 1.2s across 10+ internal modules.
 - Validated JWT, OAuth, and token-based authentication flows using Postman automation scripts, and enforced API governance through versioning, schema validation, and OpenAPI (Swagger) documentation.
 - Engineered scalable RESTful APIs using **Django REST Framework** with **PostgreSQL**, enforcing RBAC at the API layer and leveraging Django signals and middleware for audit logging and notification workflows across health benefits modules.
 - Built custom **LangChain** workflows with Prompt Templates, Chains, and ConversationBufferMemory to deliver context-aware, streaming AI interactions with consistent response quality across multi-turn conversations.
 - Built AI-native products from 0→1, integrating autonomous AI agents with **Elasticsearch** for semantic search and **Redis** for caching and session management, delivering production-ready systems from initial concept to real-world usage.
 - Led **Agile sprint planning** across cross-functional teams of 8-12 members, conducted code reviews enforcing Python and TypeScript best practices, mentored 3+ junior developers, and communicated technical tradeoffs to product.
 - Containerized services with **Docker**, deployed on **AWS (EC2, S3, RDS, API Gateway)**, and automated **CI/CD pipelines** via GitLab with environment-specific configurations, achieving 99.9% deployment reliability and 40% faster release cycles.
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Rapid Innovation

Software Developer

Noida, India

Feb 2022 – Jun 2023

- Took ownership of **code review practices** for the frontend team, introduced Git branching conventions (feature branches, PR templates, and merge policies), and mentored junior developers on **SOLID principles, component composition, state management patterns, and writing testable code**, which brought down integration conflicts and raised overall code quality.
 - Wrote pytest test suites covering **Django views, ORM queries**, serializers, and API contract validation, maintaining high coverage across critical backend services and catching regressions before they reached production.
 - Built full-stack features for a **financial analytics platform** using **React.js, TypeScript, Python Django, GraphQL, PostgreSQL, and Redis**, delivering **25+ dashboards and report views** with drilldown navigation, complex filtering, and aggregation.
 - Improved application performance by 35% by profiling React component trees, eliminating unnecessary re-renders, applying **RTK Query** caching, **lazy loading**, and code splitting, cutting dashboard load time from 8 to under 2 seconds.
 - Used Django signals to trigger async post-save workflows for notifications, audit logging, and cache invalidation, decoupling side effects from core business logic and improving maintainability across the codebase.
 - Set up **GitHub Actions CI/CD pipelines** for automated builds and **AWS** infrastructure using **Terraform**, added **Pytest, Jest React Testing**, test coverage across frontend and backend code, reduced deployment friction by **15%** across sprint cycles.
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Multiplier AI Solutions

Software Developer

Hyderabad, India

Apr 2020 – Jan 2022

- Used **Python Django** signals to trigger async post-save workflows for notifications, audit logging, and cache invalidation, decoupling side effects from core business logic and improving maintainability across the codebase.
- Designed a **centralized Doctors CRM** using on **Django ORM, React component design, JavaScript performance patterns, and REST API integration best practices**, improving overall code consistency across the codebase.

- Debugged complex UI issues across high-interaction workflow screens by systematically tracing problems through **React component state, event handlers, and Python Django REST API** responses, applied **debouncing, throttling, and state optimization** to fix the root causes, and improved UI responsiveness by **20%** across the screens staff used most.
- Built a large-scale healthcare data pipeline using Python and automation scripts to collect, normalize, and enrich records for **70,000+ physicians** across LinkedIn, Google, and medical sources with CAPTCHA handling and scheduled refresh cycles.
- Built Django middleware components for centralized request logging, **JWT token** validation, and rate limiting, keeping cross-cutting concerns cleanly separated from business logic across all API endpoints.
- Owned bug triage with QA and business stakeholders, consistently investigated root causes rather than applying surface patches, documented findings clearly for non-technical stakeholders, and kept the same classes of issues.

Key Projects

HR Management Web Platform | Perky *React.js, Next.js, TypeScript, Python Django, Gen AI, AWS (EC2, RDS, S3), Docker, LLM*
HR and benefits platform where employees and managers manage leave balances, health benefits, and approval workflows through role-based dashboards. Integrated an LLM-powered policy assistant using LangChain, OpenAI, and ChromaDB RAG pipelines to answer employee queries with contextual accuracy, reducing HR support inquiries significantly. Built for multi-tenant scale with strict RBAC, audit trails, and compliance requirements. Deployed on AWS with a Dockerized CI/CD pipeline across three environments for consistent and reliable releases.

Financial Analytics Web Application | Rapid Innovation *React.js, TypeScript, Python Django, SQL, Redis,, AWS*
Reporting platform for finance and business teams to track metrics, run drilldowns, and generate reports across large datasets in real time. Built 25+ dashboards through a centralized component system, with Redis caching and optimized query pipelines cutting load times from 8 seconds to under 2 seconds under heavy concurrent usage. Deployed on AWS with automated testing and CI/CD pipelines for predictable releases.

Medicine E-Commerce Platform | Multiplier AI Solutions *React.js, Python Django, Stripe, AWS (EC2, S3, RDS), PostgreSQL*
Online medicine selling platform where customers could browse, search, and purchase medicines with real-time inventory tracking and secure checkout. Integrated Stripe for end-to-end payment processing including one-time purchases and prescription-based order flows. Deployed on AWS with S3 for product image storage, RDS for order and inventory data, and EC2 for compute, handling reliable transactions at scale across a growing user base.

Technical Skills

- **Frontend:** React.js, Next.js (App Router, SSR/SSG), TypeScript, JavaScript, Redux Toolkit, Zustand, RTK Query, React Hooks, Context API, Tailwind CSS, Sass, HTML5, CSS3, WCAG 2.1 Accessibility, Responsive Design
- **Backend:** Python, Django, Node.js, Express.js, FastAPI, REST APIs, GraphQL, Apollo, Terraform, JWT Authentication, RBAC, Session Validation, API Contract Design
- **AI / LLM Engineering :** Langchain, RAG, Agentic AI, Langsmith, LangGraph, Open AI Models, Ollama, Hugging Face
- **Document Intelligence :** PDF/TXT/DOCX loaders, textchunking, Embeddings, ChromaDB, Pinecone
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis, SQL Server, Vector DB
- **Testing:** Jest, React Testing Library, pytest, Integration Testing, Regression Testing, Unit Testing
- **DevOps / Tools :** GitLab CI/CD, Kubernetes, Helm, Docker, Pipeline Automation, pytest, SRE Practices, Health Monitoring, Horizontal Pod Autoscaling

Education

Master of Science in Computer Science Avila University GPA: 3.63	Dec 2024
Bachelor of Engineering in Computer Engineering Kakinada Institute of Engineering and Technology GPA: 3.58	Mar 2020

Certifications & Awards

- AWS Certified Solutions Architect, 2021
- NVIDIA AI RAG Developer Contest Winner, 2024